

Andrew P. Grau

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Education

Georgia Institute of Technology | Atlanta, GA

Bachelor of Science in Electrical Engineering, GPA 3.47

August 2025 – Present

Expected Graduation May 2029

Georgia Tech Europe | Metz, France

May 2026 – August 2026

Study abroad program based in eastern France (11 credit hours)

Skills

Programming: C, Assembly, MATLAB, Python, Verilog/SystemVerilog/VHDL

Hardware: 3D Printers, Single Board Computers: Raspberry Pi, Microcontrollers: ESP32, Arduino

Software: LTspice, Quartus, KiCad, Onshape, BambuStudio, PluraPrint, SimVision, Cadence Xcelium/Innovus/Genus/IMC

Lab & Instrumentation: Cleanroom certified (Marcus Nanotechnology Building), TPT HB16 Leica S9D wire bonder, Tektronix 370B curve tracer, probe station + SMU, oscilloscope, DMM

Communication: Design proposals, funding pitches, presentations (large and small audiences)

Experience

Georgia Tech Vertically Integrated Projects Program | Atlanta, GA

January 2026 – Present

Student Researcher, Team Lead (August 2026) / Compound Semiconductor Products Team

Design, simulate, fabricate, and characterize custom compound semiconductor devices and circuits (e.g. GaAs-, GaN-, or InP-based). Partnered with Sandia National Laboratories for mentorship on EDA/PDK tools and applications and access to national laboratory resources for design, simulation, fabrication, test and characterization. US Persons only (ITAR and other export control/access restrictions).

- Characterized 5 GaN HEMT samples from Sandia National Labs using a Tektronix 370B curve tracer to quantify device-to-device variability and identified and corrected hysteresis artifacts in the I-V sweeps
- Designing op-amp circuit in LTspice using GaN HEMTs for fabrication and testing through partnership with Sandia National Labs
- Lead a 23-person research team: run meetings, manage subteam assignments and logistics, and onboard new members onto tooling and study material

Georgia Tech-CNRS International Research Laboratory (IRL 2958) | Metz, France

May 2026 – August 2026

Undergraduate Research Assistant, advised by Dr. Ashutosh Srivastava

- Conducted exploratory CO₂ gas-sensing experiments on undoped and doped h-BN, benchmarked against GaN and Si controls, to assess h-BN's viability as a sensing material
- Measured current-vs-time response on a probe station with an SMU under controlled CO₂ delivery (8–10 psi, ~0.5–1 cm standoff), collecting data in Clarius against a defined criterion: sustained current shift in h-BN with no response from controls
- Authored a technical report covering the sensing study and prior device characterization; results were preliminary and motivated follow-on testing under temperature regulation

Georgia Institute of Technology | Atlanta, GA

October 2025 – Present

Student Assistant / ECE CommX Program

- Co-planning an ECE department project competition funded by an earmarked donation: ~10 student teams receive build stipends and present at a spring showcase for prize funding
- Proposed and drove ECE 3005 course adoption of BuzzCard scanning to replace handwritten attendance cards for 500+ students, eliminating manual card entry, restocking, and improving classroom efficiency
- Designing an ECE 3005-wide census to collect career and professional outcome data on ECE undergraduates
- Served as substitute TA for a 100-student section of ECE 3005 (Professional and Technical Communications); resolved student grade disputes and managed CommX Studio operations

Interdisciplinary Makerspace @ Georgia Tech | Atlanta, GA

August 2026 – Present

Peer Instructor (PI) / The Hive

Largest interdisciplinary makerspace in the country catering specifically to electronics and computer engineering design and innovation. Providing hands-on learning environment in six technical areas: PCB fabrication, embedded systems, electronics benchtop, 3D printing, laser cutting/engraving, and woodworking and metalworking. Serving 5,000 users per month.

- Lead tours of 15,000+ sq ft facility for end-users and prospective students; enforce makerspace policies and manage end-user check-in
- Run the FDM 3D print room (4,000+ prints/year), using Bambu Lab P1S and H2S/H2D printers, managing the queue in PluraPrint and diagnosing failed prints, slicer settings, and hardware failures
- Support end-users through project builds across the space's six technical areas, from design questions through fabrication

SiliconJackets | Atlanta, GA

March 2026 – Present

Analog Subteam (currently onboarding), Previously Physical Design / Student Organization

- Implemented a 64-bit addition calculator in Verilog/SystemVerilog against a provided harness, extending the controller FSM with two added states to handle operand read timing and pointer increment; verified in Xcelium and debugged in SimVision
- Extended a UVM testbench (driver, monitor, scoreboard, sequence items) with SVA assertions covering reset behavior, carry propagation, and read/write protection, reaching 97.5% coverage in Cadence IMC
- Ran the RTL2GDS flow from Genus synthesis through Innovus place-and-route; closed setup timing by adjusting the SDC clock period and completed Tcl procs automating macro, timing-path, and clock-tree fanout inspection
- Root-caused 3,500+ DRC violations around ChipFoundry SRAM macros to the router landing met1 vias inside the macro's OBS blockage; resolved by lowering the bottom routing layer to met1 and adding a placement halo, producing a DRC-clean design

Projects

Affordable Photolithography Machine | Personal Project

March 2026 – Present

Independent work; enrolled as ECE 4699 Special Topics (Fall 2026), advised by Dr. William Hunt

Designing a desktop maskless photolithography stepper targeting 5µm critical dimension, as a low-cost alternative to \$200K+ commercial photolithography systems for university and community-college labs.

- Conducted 10+ customer discovery calls with researchers at Penn State, UPenn, Washington, Dartmouth, RIT, UIUC, UCSD, Georgia Tech, and Sandia National Laboratories to define target specs and price point
- Prepared project pitch deck and presented to Jonathan Goldman, Director of Quadrant-i (Georgia Tech's research commercialization program)
- Secured \$1,700 in project funding; full system BOM under \$2,000 against \$200K+ commercial equivalents
- Designed complete optomechanical system in CAD: custom plano-convex collimator module converting the projector's diverging output to a converging pattern, reverse-fed into a repurposed compound microscope with 4x/10x/40x objectives
- Built XY stage using dual NEMA11 steppers with debounced limit switches and homing, driven by a TB6600 driver and Arduino Mega over a serial command interface
- Integrated Raspberry Pi 5 as system controller for stage commands, camera feed, and projector HDMI pattern output; 24V 25A supply for the electronics
- Specified and validated custom machine control software for multi-layer wafer layout, GDSII/DXF import, and exposure job sequencing
- Achieved ~50µm critical dimension with a visible-light 3LCD test source; UV cleanroom exposures on the 40x objective in progress to close the gap to target

F1 LED Matrix Display with Buck Converter | Personal Project

August 2026 – Present

Wall-mounted 96x144 LED Matrix display for F1 featuring live race updates, championship standings, and countdown to next Grand Prix.

- Designed a buck converter around the LT1767EMS8-3.3 switching regulator so a single 5V supply feeds both the three LED panels and the 3.3V ESP32-S3, eliminating a second rail
- Verified the topology in LTspice for output regulation and ripple (1mV peak-to-peak) before committing to hardware
- Laid out a custom PCB in KiCad with custom silkscreen artwork
- Modeled the enclosure and internal support structure in Onshape for laser cutting
- Specified and integrated a live data pipeline over the OpenF1 REST API: race order, intervals, qualifying times, and both championship tables
- Defined a time-based FSM that switches the display between qualifying, race, championship, and off-season modes from the session schedule, with no user input

Awards

Coca-Cola Scholar | Coca-Cola Scholarship Foundation

April 2025

Eagle Scout | Scouting America

March 2025